TTK4190 Assignment 2 Part 1
Marine Craft Modeling

Objective
This is the first part of a three-part assignment where you will create a high-fidelity simulation of a ship. The simulation model and its functionality will be built up step-by-step through these assignments. For the simulation problems, you will use the Matlab template files `ship.m` and `main.m` provided on Blackboard. To run these templates, the MSS toolbox must be installed. The easiest way to do this is to install it as an Add-On, `https://se.mathworks.com/matlabcentral/fileexchange/86393-marine-systems-simulator-mss`.

In this assignment, you will go through and implement the hydrodynamic concepts introduced in the course.

Delivery Details
The tasks may be completed individually or in groups. The hand-in should be a PDF report and the code (.m files) for each sub-problem. Upload the .m files directly, not in a zip folder.

To simplify the analysis in problems 1 and 2, we assume the geometrical shape of the ship to be similar to a rectangular prism (see Figure 1). Furthermore, the relevant theory for this assignment is located in Chapters 3, 4, 5, and 6 (Fossen, 2021).

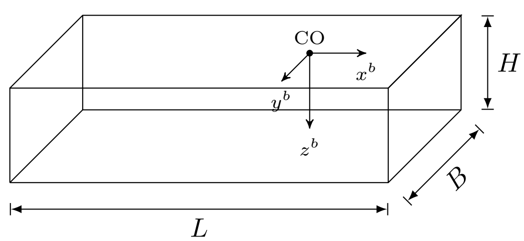

Figure 1: A rectangular prism with length, beam, and height dimensions $L = 161$ m, $B = 21.8$ m, and $H = 15.8$ m, respectively. The coordinate origin of the body fixed reference frame $\{b\}$ is denoted CO.

Problem 1: Rigid-Body Kinetics of a Rectangular Prism

One of the first steps in creating a dynamic ship model is determining the rigid-body mass matrix $\mathbf{M}_{\mathrm{RB}}$ and the Coriolis matrix $\mathbf{C}_{\mathrm{RB}}$. In this problem, you will find these matrices based on the rectangular prism parameters in Figure 1. Parameters for a real ship of the similar size as the prism are given below as reference.

```
% real ship parameters
rho = 1025;             % density of water (kg/m^3)
m = 17.0677e6;          % mass (kg)
Iz = 2.1732e10;         % yaw moment of inertia about CO (kg m^3)
xg = -3.7;              % x-coordinate  of CG relative CO (m)
L = 161;                % length (m)
B = 21.8;               % beam (m)
T = 8.9;                % draft (m)
% added mass about CO
Xudot = -8.9830e5;
Yvdot = -5.1996e6;
Yrdot =  9.3677e5;
Nvdot =  Yrdot;
Nrdot = -2.4283e10;
```

(a) The mass of the rectangular prism in Figure 1 is given by $m = \rho_m LBH$, where ρ_m denotes the density of the body. Find an expression for the moment of inertia I_z^{CG} about the prism's center of gravity (CG) and compute the numerical value. Assume the prism and ship have the same mass. Show/explain why $I_{xy}^{CG} = I_{yx}^{CG} = 0$. *Hint:* The CG is located in the volume center of the body, and the mass is homogeneously distributed.

(b) Let $\mathbf{r}_{bg}^b = [x_g = -3.7,\ y_g = 0,\ z_g = H/2]^\top$ denote the distance from the coordinate origin CO on the body to CG. Find an expression for I_z^{CO} about CO for the rectangular prism and compute the numerical value. What is the ratio between the moments of inertia of the prism and the real ship?

(c) Write down analytical expressions for $\mathbf{M}_{\mathrm{RB}}$ and $\mathbf{C}_{\mathrm{RB}}$ about the CO using $\mathbf{r}_{bg}^b$ from (b). This should be done for surge, sway, and yaw (3 DOFs) only. *Hint:* See eqs. (3.29,3.30) in (Fossen,2021).

(d) Verify that $\mathbf{C}_{\mathrm{RB}}$ is skew-symmetric. Why is this a desired property?

(e) The Coriolis matrix $\mathbf{C}_{\mathrm{RB}}^{\mathrm{CO}}$ above possesses another property that is useful when irrotational ocean currents enter the equations of motion. Can you explain this property?

Problem 2: Hydrostatics of a Rectangular Prism

In this task, we are going to determine some important hydrostatic quantities using the rectangular prism in Figure 1. Let ∇ denote the displaced water volume caused by the prism, ρ the water density, and g the gravitational constant. In this problem (problem 2), let the draft of the prism be $\boldsymbol{T = 4.74}$ **m** and assume that the roll and pitch angles of the prism are negligible.

(a) Compute the volume displacement ∇.

(b) Compute the waterplane area A_{wp} and write an expression for the hydrostatic force Z_{hs} in heave.

(c) Compute the period in heave using eq. (4.78) in (Fossen, 2021).

(d) Find analytical expressions for GM_T and GM_L and compute the numerical values.

(e) Using the numerical values of GM_T and GM_L, explain whether the prism is metacentrically stable or not.

Problem 3: Added Mass and Coriolis

Obtaining accurate estimates of the added mass coefficients is not straightforward. The coefficients are usually computed numerically using hydrodynamic potential theory programs such as ShipX or WAMIT, which integrate fluid pressures over the surface hull for different excitation frequencies. A less accurate approach is to use semi-empirical methods to compute the added mass derivatives. Whichever approach is taken, finding the added mass coefficients, analytically or numerically, is a demanding task and, therefore, outside this course's scope.

In the following problems, consider the general 6×6 matrix of added mass coefficients. Assume no couplings between the surge, heave-roll-pitch, and sway-yaw subsystems.

$$\mathbf{M}_A = \mathbf{M}_A^\top = -\begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix} > 0 \tag{1}$$

(a) Rewrite (1) into a 3×3 matrix consisting of surge, sway, and yaw components only.

(b) Find the Coriolis force matrix $\mathbf{C}_A(\boldsymbol{\nu})$ due to added mass. The matrix dimensions should be 3×3 with components corresponding to surge, sway, and yaw. *Hint:* see *Property 6.2* in (Fossen, 2021).

Problem 4: Implementing the Maneuvering Model in Matlab

In this task, we are going to combine results from the previous problems to create a complete simulation model in Matlab. Download the templates `ship.m` and `main.m` from Blackboard. Don't worry about the ship acting weird in the simulation. In this exercise, you will make several changes to the simulation code to make the ship's movement (dynamics) realistic.

(a) The rigid-body matrices $\mathbf{M}_{\mathrm{RB}}$ and $\mathbf{C}_{\mathrm{RB}}$ have already been implemented in the code. Numerical values for added mass are also included. Your task in this problem is to add the matrices $\mathbf{M}_{\mathrm{A}}$ and $\mathbf{C}_{\mathrm{A}}$ to `ship.m` and test that your code runs with the hydrodynamic added mass effects.

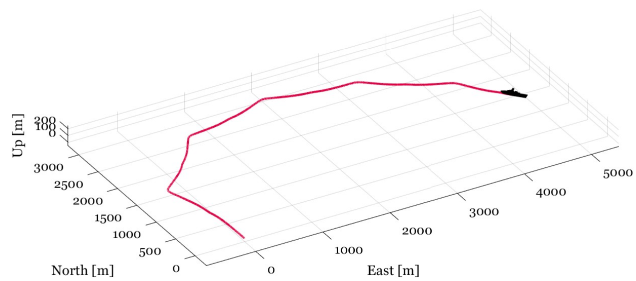

Figure 2: *Hint:* result after implementing 4a.

(b) Add a *diagonal* linear damping matrix $\mathbf{D}$ with

$$-X_u = \frac{m - X_{\dot{u}}}{T_1}, \qquad -Y_v = \frac{m - Y_{\dot{v}}}{T_2}, \qquad -N_r = \frac{I_z - N_{\dot{r}}}{T_6}$$

where $T_1 = T_2 = 20$ s and $T_6 = 10$ s are the natural periods in surge, sway, and yaw as described in Ch. 4.3 in (Fossen, 2021). Don't confuse these with the time constants in the Nomoto model (Ch. 7.2.1, Fossen, 2021).

After adding linear damping, add nonlinear surge damping described in eqs. (6.76,6.77) and cross-flow drag in eqs. (6.87,6.88) in (Fossen, 2021). Assume that $k = 0.1$, $C_R = 0$, and $\epsilon = 0.001$ (a small number to ensure that C_f is well-defined at $u_r = 0$). The integrals can be coded as a for loop with the draft $T(x) = T =$ constant and $C_d^{2D}(x) =$ constant, given by `Hoerner.m` function in the MSS toolbox. The for loop is already given to you below, copy it over to your code in `ship.m`; then you just have to understand what the loop computes and how to use its' output.

```
% Strip theory: cross-flow drag integrals
dx = L/10;                                               % 10 strips
for xL = -L/2:dx:L/2
    Ucf = abs(v_r + xL * r) * (v_r + xL * r);
    Ycf = Ycf - 0.5 * rho * T * Cd_2D * Ucf * dx;        % sway force
    Ncf = Ncf - 0.5 * rho * T * Cd_2D * xL * Ucf * dx;   % yaw moment
end
```

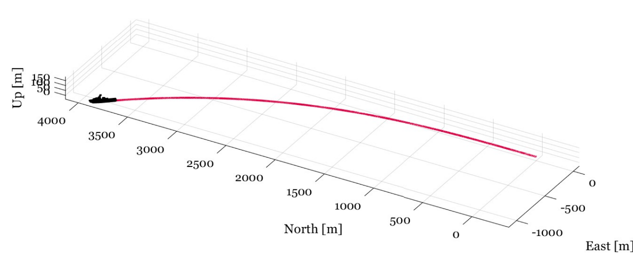

Figure 3: *Hint:* result after implementing 4b.

(c) Implement time-varying input commands for rudder deflection and propeller speed. Using the time-varying inputs, design two step-response maneuvers, one with a step change in rudder position and one with a step change in the propeller speed. The changes should be made in `main.m` where you have the simulation loop. In the report, attach plots for each of the step-response maneuvers.

(d) You are given the rigid-body equations of motion for a vessel, without added mass effects, as presented in equation (3.47) of Fossen (2021):

$$\boldsymbol{M}_{RB}\,\dot{\boldsymbol{\nu}} + \boldsymbol{C}_{RB}(\boldsymbol{\nu})\,\boldsymbol{\nu} = \boldsymbol{\tau}_{RB}$$

where $\boldsymbol{\nu}$ is the vessel velocity state vector in the body frame, and $\boldsymbol{\tau}_{RB}$ is the vector of external forces and moments acting on the vessel. Explain how this equation can be used to simulate the vessel dynamics. Write pseudocode or provide a flow chart that outlines the steps needed to propagate the vessel state over time.

Hint: this is already implemented in the code handed out to you.

Selected numerical answers

1a) $I_z^{CG} = 3.7544\text{e}{+}10$

1b) $I_{z,\text{prism}}^{\text{CO}} = 3.7777\text{e}{+}10$, $I_{z,\text{prism}}^{\text{CO}}/I_{z,\text{boat}}^{\text{CO}} = 1.7383$

2c) $T_3 \approx 6.1766$

2d) $GM_T = 2.8251$, $GM_L = 450.1838$